University of Florida, Department of Mathematics

Analysis PhD Exam, January 1988

Answer all six questions. Justify all work. To obtain any partial credit, all work must be presented in a neat and logical fashion.

1. State and prove the Radon–Nikodym Theorem.

2. State and prove the Hahn–Banach Theorem.

3. Let Ω be a topological space, $\mathcal{B}(\Omega)$ the Borel subsets of Ω. Suppose that for each n, μ_n is a countably additive positive finite regular measure defined on $\mathcal{B}(\Omega)$. Let

$$\mu(\cdot) = \sum_{n=1}^{\infty} \frac{\mu_n(\cdot)}{2^n\big(1+\mu_n(\Omega)\big)}.$$

Show that μ is also a regular countably additive measure.

4. Let $\{f_n : n \geq 0\}$ and $\{g_n : n \geq -1\}$ be two sequences of real-valued measurable functions on a measurable space $(\Omega, \mathcal{F})$. Let $B \subset \mathbb{R}^1$ be a Borel set on $\mathbb{R}^1$. Define

$$T(w) = \begin{cases} \sup\{n \geq 0 : f_n(w) \in B\}, & \text{if } \{n \geq 0 : f_n(w) \in B\} \text{ is bounded and nonempty,} \\ 0, & \text{if } \{n \geq 0 : f_n(w) \in B\} = \varnothing, \\ -1, & \text{if } \{n \geq 0 : f_n(w) \in B\} \text{ is unbounded.} \end{cases}$$

(A) Prove T is $\mathcal{F}$-measurable.

(B) Define $h(w) = g_{T(w)}(w)$; prove h is $\mathcal{F}$-measurable.

5. Let $T : (-\pi/2, \pi/2) \to \mathbb{R}$ be given by $T(x) = \tan x$. Let λ be Lebesgue measure on $(-\pi/2, \pi/2)$ and define a measure μ on $(\mathbb{R}, \mathcal{B}(\mathbb{R}))$ by setting $\mu(A) = \lambda(T^{-1}(A))$. Compute $\dfrac{d\mu}{dx}$.

6. Warning: Suspected error. Problem 6 uses q without defining it. Also, the retained Hint 2, $g = |f|^{p-1} \in L^q$, does not account for the sign of a general f, and the endpoint $p = 1$ requires separate treatment. The intended correction is not uniquely determined from the immediate context.

Fix $1 \leq p < \infty$, and let $f \in L^p(X, \mathcal{A}, \mu)$. Show that

$$\|f\|_p = \sup\left\{\int fg\,d\mu : \|g\|_q \leq 1\right\}.$$

Hint:

1. Show that if $\|g\|_q \leq 1$, then $\int fg\,d\mu \leq \|f\|_p$.
2. Show that $g = |f|^{p-1} \in L^q$.
3. State why doing 1. and 2. solves the problem.